Review & Examples Wednesday Lecture.

The Inertia Tensor

- Properties of Rigid Objects
- $I_{/a}$ in 2D
  - ↳ Example
- $\underline{\underline{I}}_{/a}$ in 3D

HW5 due Friday.

Prelim 1 10/6-7

Review

Mass: CoG $\quad m\vec{r}_a = \int \vec{r}\, dm$

Linear Momentum: $\begin{cases} \vec{p} = \int \vec{v}\, dm = m\vec{v}_a \\ \dot{\vec{p}} = m\vec{a}_a \end{cases}$

Angular Momentum: $\vec{H}_{/I} = \int \vec{r}_{/I} \times \vec{v}_I \, dm$

$$= \vec{H}_{a/I} + \vec{H}_{/a}$$

Angular Velocity $\quad \vec{\omega} \times \vec{r} = \vec{v}$

$$\Rightarrow \vec{H}_{/a} = \int \vec{r}_a \times (\vec{\omega} \times \vec{r}_a)\, dm$$

in 2D $\quad = \omega \underbrace{\int \vec{r}_a^{\,2}\, dm}_{\text{Polar Moment of Inertia.}} \hat{k}$

2D Inertia Tensor $\underline{\underline{I}}_G$

Hint

$$\boxed{\vec{a}\times(\vec{b}\times\vec{c}) = \vec{b}(\vec{a}\cdot\vec{c}) - \vec{c}(\vec{a}\cdot\vec{b})}$$

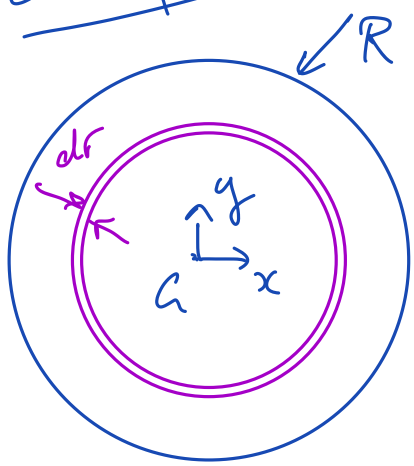
$$\vec{H}_{/G} = \int \vec{r}_G \times (\vec{\omega}\times\vec{r}_G)\, dm$$

$$= \int \vec{\omega}\underbrace{(\vec{r}\cdot\vec{r})}_{r^2} - \vec{r}\,\cancel{(\vec{r}\cdot\vec{\omega})}^{\,0}\, dm$$

$$\vec{r} = \begin{bmatrix} x\hat{i} \\ y\hat{j} \end{bmatrix} \qquad \vec{\omega} = \omega\hat{k}$$

$$= \omega\underbrace{\int r^2\, dm}_{\underline{\underline{I}}_G}\,\hat{k} = \omega\underbrace{\int (x^2+y^2)\, dm}_{\underline{\underline{I}}_G}\,\hat{k}$$

Example Disk



$$I_G = \int r^2\, dm \qquad dm = \rho\, 2\pi r\, dr$$

3D Inertia Tensor

$$\vec{r} = \begin{bmatrix} x\hat{i} \\ y\hat{j} \\ z\hat{k} \end{bmatrix} \qquad \vec{\omega} = \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$$\vec{H}_a = \int \vec{r}_a \times (\vec{\omega} \times \vec{r}_a)\, dm$$

$$= \int \vec{\omega}(\vec{r}\cdot\vec{r}) - \vec{r}(\vec{\omega}\cdot\vec{r})\, dm$$

$$= \int \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}(x^2+y^2+z^2) - \begin{bmatrix} x\hat{i} \\ y\hat{j} \\ z\hat{k} \end{bmatrix}(\omega_x x + \omega_y y + \omega_z z).\, dm$$

$$\vec{H}_a \hat{i} = \int \omega_x(\cancel{x^2}+y^2+z^2) - \cancel{x^2\omega_x} - xy\omega_y - xz\omega_z.\, dm$$

$$= \int \omega_x(y^2+z^2) - \omega_y xy - \omega_z xz\ dm\ \hat{i}$$

$$\vec{H}_a \hat{j} = \int \omega_y(x^2+\cancel{y^2}+z^2) - yx\omega_x - \cancel{y^2}\omega_y - yz\omega_z\ dm$$

$$= \int \omega_y(x^2+z^2) - \omega_x yx - \omega yz.\ dm\ \hat{j}$$

$$\vec{H}_a \hat{k} = \int \omega_z(x^2+y^2+\cancel{z^2}) - zx\omega_x - zy\omega_y - \cancel{z^2}\omega_z\ dm$$

$$= \int \omega_z(x^2+y^2) - \omega_x zx - \omega_y zy\ dm\ \hat{k}$$

$$\vec{H}_{/a} = \underline{\underline{I}}_{/a}\vec{\omega} = \begin{bmatrix} \int y^2+z^2 \cdot dm & \int -xy\, dm & \int -xz \cdot dm \\ \int -yx \cdot dm & \int x^2+z^2 \cdot dm & \int -yz \cdot dm \\ \int -zx \cdot dm & \int -zy\, dm & \int x^2+y^2 \cdot dm \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix}$$

$$\underline{\underline{I}}_a = \begin{bmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{bmatrix}$$

Products of Inertia

Moments of Inertia.

Alternatively

$$\vec{H}_a = \int \vec{r}_{/a}^{\,2}\, \underline{\underline{1}} - \vec{r}_{/a} \otimes \vec{r}_{/a} \cdot dm$$

3x3 Identity Matrix

Try yourself!